

Supplementary Material for “Canonical coupled cluster binding benchmark for nanoscale noncovalent complexes at the hundred-atom scale”

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TABLE S1: Binding correlation energies (in kcal/mol) for the coronene dimer (C2C2PD) and stacked guanine–cytosine pair (GCGC) in L7 obtained at the DLPNO-CCSD(T₁)/def2-TZVPP level with TightPNO and VeryTightPNO thresholds. The absolute energy change, $|\Delta E|$, resulting from tightening the threshold from TightPNO to VeryTightPNO is also shown.

	DLPNO-CCSD(T ₁) TightPNO	DLPNO-CCSD(T ₁) VeryTightPNO	$ \Delta E $
C2C2PD	−35.70	−33.49	2.21
GCGC	−23.35	−22.72	0.63

TABLE S2: Basis-set correction term in kcal/mol, $\Delta\Delta E_{\text{corr}}^{\text{CCSD(T)}} = (E_{\text{corr}}^{\text{DLPNO-CCSD(T}_1\text{)/def2-TZVPP}} - E_{\text{corr}}^{\text{MP2/def2-TZVPP}}) - (E_{\text{corr}}^{\text{DLPNO-CCSD(T}_1\text{)/def2-SVP}} - E_{\text{corr}}^{\text{MP2/def2-SVP}})$, for the coronene dimer (C2C2PD) and stacked guanine–cytosine pair (GCGC) in L7 with TightPNO and VeryTightPNO thresholds. The absolute energy change, $|\Delta E|$, resulting from tightening the threshold from TightPNO to VeryTightPNO is also shown.

	DLPNO-CCSD(T ₁) TightPNO	DLPNO-CCSD(T ₁) VeryTightPNO	$ \Delta E $
C2C2PD	2.98	3.02	0.04
GCGC	1.73	1.70	0.03

TABLE S3: Binding correlation energies (kcal/mol) for the L14 dataset, calculated using canonical CCSD(T)/def2-SVP and various levels of DLPNO-CCSD(T)/def2-SVP. Mean absolute error (MAE) and maximum absolute error (MAX) relative to canonical CCSD(T) are also provided.

L14	canonical CCSD(T)	DLPNO-CCSD(T ₀) TightPNO	DLPNO-CCSD(T ₀) VeryTightPNO	DLPNO-CCSD(T ₁) TightPNO	DLPNO-CCSD(T ₁) VeryTightPNO
C2C2PD	−27.42	−28.55	−26.33	−28.95	−26.77
C3A	−18.22	−19.68	−17.72	−19.90	−17.99
C3GC	−31.91	−33.97	−31.05	−34.35	−31.52
CBH	−12.36	−12.38	−11.98	−12.45	−12.07
GCGC	−16.81	−16.97	−16.28	−17.13	−16.47
GGG	−7.14	−7.24	−6.99	−7.31	−7.08
PHE	−4.26	−4.21	−3.99	−4.23	−4.04
2a	−28.55	−29.58	−27.49	−29.90	−27.90
3b	−27.01	−28.11	−26.25	−28.35	−26.57
2b	−22.57	−23.44	−21.90	−23.74	−22.24
S8-1	−27.32	−27.86	−26.41	−28.31	−26.93
S8-2	−23.26	−23.95	−22.37	−24.22	−22.71
Da1	−29.08	−30.01	−28.14	−30.40	−28.58
Da2	−26.28	−27.26	−25.46	−27.59	−25.84
MAE	—	0.79	0.70	1.05	0.39
MAX	—	2.06	1.09	2.44	0.65

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TABLE S4: Percentage contribution of (T) to the total CCSD(T) correlation binding interaction energies for 14 complexes in the L14 dataset, as well as the parallel-displaced benzene dimer from the S66 dataset, computed at the canonical CCSD(T)/def2-SVP level.

	(T) contribution
parallel-displaced benzene dimer	15.3%
C2C2PD	18.1%
C3A	18.0%
C3GC	18.0%
CBH	13.8%
GCGC	16.1%
GGG	17.4%
PHE	24.5%
2a	18.5%
2b	17.6%
3b	17.2%
S8-1	19.7%
S8-2	18.7%
Da1	17.6%
Da2	17.4%

TABLE S5: Canonical CCSD(cT)-fit/CBS binding interaction energies (kcal/mol) for 14 complexes in the L14 dataset, calculated using the (T)/X approach to estimate the (cT) contribution from (T) in all three CCSD(T) terms of the composite scheme. Here, $X = a + b \cdot Y$, where Y represents the ratio of MP2 to CCSD correlation contributions, and $a = 0.7764$ and $b = 0.2780$, as proposed by Tobias Schäfer *et al.*, arXiv:2407.01442 (2024). The CCSD(cT) method predicts weaker binding interactions compared to CCSD(T), and the binding difference in kcal/mol, ΔE , is also reported.

L14	canonical CCSD(cT)-fit/CBS	ΔE^a
C2C2PD	-19.13	1.52
C3A	-15.38	0.97
C3GC	-26.53	1.57
CBH	-10.78	0.28
GCGC	-13.13	0.64
GGG	-1.92	0.29
PHE	-24.90	0.26
2a	-32.65	1.49
2b	-20.95	1.08
3b	-31.78	1.21
S8-1	-34.91	1.63
S8-2	-29.53	1.26
Da1	-17.70	1.50
Da2	-18.88	1.32

$$^a E_{\text{int}}^{\text{CCSD(cT)/CBS}} - E_{\text{int}}^{\text{CCSD(T)/CBS}}$$

TABLE S6: DLPNO-CCSD(cT₁)-fit/CBS (VeryTightPNO) binding interaction energies (kcal/mol) for 11 complexes in the vL11 dataset, calculated using the (T)/X approach to estimate the (cT) contribution from (T) in all three CCSD(T) terms of the composite scheme. Here, $X = a + b \cdot Y$, where Y represents the ratio of MP2 to CCSD correlation contributions, and $a = 0.7764$ and $b = 0.2780$, as proposed by Tobias Schäfer *et al.*, arXiv:2407.01442 (2024). The DLPNO-CCSD(cT₁) method predicts weaker binding interactions compared to DLPNO-CCSD(T₁), and the binding difference in kcal/mol, ΔE , is also reported.

	vL11 DLPNO-CCSD(cT ₁)-fit/CBS	ΔE^a
3a	-36.38	1.63
4a	-35.69	2.83
5a	-42.02	0.71
5b	-28.78	0.73
CiM-a	-40.87	0.40
CiM-c	-14.31	0.43
CiM-d	-72.99	0.70
CiM-e	-65.20	1.18
DNA-ellipticine	-36.79	1.59
C ₆₀ -C ₆₀	-8.88	0.67
C ₆₀ @[6]CPPA	-36.53	3.66
$^a E_{\text{int}}^{\text{DLPNO-CCSD(cT}_1\text{)}/\text{CBS}} - E_{\text{int}}^{\text{DLPNO-CCSD(T}_1\text{)}/\text{CBS}}$		

TABLE S7: Binding energies and corresponding energy components (in kcal/mol) for the 14 complexes in the L14 dataset, computed using SAPT0+aiD(CCD)/aug-cc-pVTZ. The interaction character is designated as “D” (dispersion-dominated, $|E_{\text{disp}}/E_{\text{elst}}| > 2$), “E” (electrostatics-dominated, $|E_{\text{disp}}/E_{\text{elst}}| < 0.5$), or “M” (mixed character, intermediate $|E_{\text{disp}}/E_{\text{elst}}|$).

L14	E_{elst}	E_{exch}	E_{ind}	E_{disp}	E_{total}	$ E_{\text{disp}}/E_{\text{elst}} $	Type
C2C2PD	-9.84	28.49	-2.68	-37.90	-21.93	3.85	D
C3A	-8.38	19.86	-2.19	-27.81	-18.52	3.32	D
C3GC	-14.24	36.82	-5.10	-48.87	-31.39	3.43	D
CBH	-2.86	11.32	-1.18	-19.83	-12.55	6.93	D
GCGC	-8.57	23.44	-2.55	-28.52	-16.21	3.33	D
GGG	3.67	6.10	-1.61	-10.19	-2.03	2.78	D
PHE	-29.30	26.26	-10.60	-16.31	-29.95	0.56	M
2a	-31.66	53.47	-11.46	-52.46	-42.10	1.66	M
2b	-19.61	37.10	-5.97	-38.41	-26.89	1.96	M
3b	-20.67	36.33	-6.37	-46.45	-37.16	2.25	D
S8-1	-33.49	51.73	-12.36	-47.95	-42.07	1.43	M
S8-2	-25.07	39.87	-7.66	-41.62	-34.48	1.66	M
Da1	-10.59	33.90	-3.23	-38.68	-18.59	3.65	D
Da2	-11.99	31.66	-3.58	-36.74	-20.65	3.07	D

TABLE S8: Binding energies (kcal/mol) for the L14 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using MP2-based methods. All MP2 calculations, except for MP2+aiD(CCD), were extrapolated to the CBS limit, with MP2+aiD(CCD) using the aug-cc-pVTZ basis set. BSSE corrections were applied to all MP2 calculations using the counterpoise method.

L14	MP2	SNS-MP2	SCS-MP2	SCS(MI)-MP2	MP2D	SCS-MP2D	MP2+aiD(CCD)
C2C2PD	-38.07	-25.78	-26.32	-28.81	-26.50	-25.50	-21.39
C3A	-27.04	-18.98	-18.95	-21.01	-19.66	-19.15	-17.75
C3GC	-45.26	-31.88	-31.24	-34.89	-32.87	-32.09	-29.51
CBH	-11.84	-10.04	-7.60	-8.66	-11.40	-11.31	-11.92
GCGC	-18.95	-13.30	-12.22	-13.53	-13.40	-13.21	-14.94
GGG	-4.50	-2.01	-1.97	-2.11	-2.61	-2.27	-3.15
PHE	-26.45	-25.75	-22.79	-25.17	-25.52	-25.70	-23.77
2a	-49.97	—	-35.84	-40.67	-38.21	-38.34	-35.68
2b	-33.74	-24.56	-23.56	-26.34	-25.00	-24.89	-24.48
3b	-45.80	—	-33.29	-36.91	-35.93	-35.46	-32.69
S8-1	-54.62	—	-40.69	-45.06	-42.19	-41.76	-39.14
S8-2	-44.33	-34.21	-32.38	-36.30	-35.19	-34.57	-30.33
Da1	-35.47	-22.84	-23.57	-25.77	-24.81	-23.61	-17.94
Da2	-34.77	-23.21	-23.69	-26.09	-25.59	-24.63	-19.33
MAE	11.26	2.40	2.59	4.54	3.32	2.89	1.24
MAX	18.08	5.13	5.68	8.52	5.85	5.22	2.60

TABLE S9: Binding energies (kcal/mol) for the L14 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using selected DFT functionals developed by the Head-Gordon group. All DFT calculations were performed with the def2-TZVPPD basis set and applied counterpoise corrections.

	L14	B97M-V	B97M-rV	ω B97X-D	ω B97X-V	ω B97M-V
C2C2PD	-20.90	-22.36	-23.18	-21.50	-22.66	
C3A	-16.53	-17.39	-17.78	-16.90	-17.90	
C3GC	-29.15	-30.44	-30.07	-29.11	-31.15	
CBH	-11.79	-12.35	-13.72	-12.04	-11.43	
GCGC	-14.39	-15.24	-14.50	-14.58	-15.18	
GGG	-1.88	-2.24	-1.91	-2.09	-2.18	
PHE	-25.10	-25.54	-25.36	-25.58	-25.80	
2a	-34.55	-35.96	-38.63	-38.61	-38.61	
2b	-22.50	-23.53	-25.43	-25.11	-25.11	
3b	-34.56	-36.15	-31.73	-35.55	-36.58	
S8-1	-37.20	-38.47	-38.79	-37.69	-39.67	
S8-2	-31.59	-32.91	-31.16	-32.01	-33.38	
Da1	-20.11	-21.28	-21.35	-20.08	-21.73	
Da2	-21.23	-22.34	-23.09	-21.56	-23.07	
MAE	0.65	1.64	1.90	1.39	2.24	
MAX	1.57	3.16	4.48	4.47	4.47	

TABLE S10: Binding energies (kcal/mol) for the L14 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using selected Minnesota functionals and functionals with D4 and MBD dispersion corrections. All DFT calculations were performed with the def2-TZVPPD basis set and applied counterpoise corrections. The three-body Axilrod–Teller–Muto effects were accounted for in the D4 dispersion correction but not in D3 in this work.

	L14	B3LYP+D4	PBE0+D4	PBE0+MBD	PBE+D4	M06L-D3	M06-2X-D3	MN15
C2C2PD	-23.99	-20.42	-18.31	-19.55	-18.06	-19.84	-18.66	
C3A	-18.70	-15.91	-14.33	-15.24	-14.04	-15.40	-13.34	
C3GC	-32.05	-26.73	-24.35	-25.63	-25.68	-27.70	-24.41	
CBH	-11.68	-11.38	-13.68	-12.32	-10.53	-7.69	-9.87	
GCGC	-16.69	-13.63	-12.16	-13.69	-13.38	-13.75	-12.44	
GGG	-2.68	-1.92	-1.55	-2.21	-1.21	-1.33	-0.92	
PHE	-25.51	-25.29	-26.05	-24.39	-24.55	-25.43	-23.15	
2a	-36.62	-34.62	-35.06	-32.36	-33.84	-36.15	-34.54	
2b	-24.02	-22.59	-22.63	-21.16	-21.64	-23.76	-22.74	
3b	-35.70	-31.01	-28.84	-30.01	-32.53	-33.66	-32.67	
S8-1	-40.69	-37.45	-36.43	-36.75	-35.23	-37.96	-35.46	
S8-2	-33.34	-29.98	-28.72	-29.30	-29.84	-31.74	-29.66	
Da1	-21.69	-18.21	-16.87	-17.68	-18.81	-19.33	-17.96	
Da2	-21.83	-19.35	-18.93	-18.49	-20.35	-21.16	-20.27	
MAE	2.29	0.68	1.81	1.24	0.99	1.04	1.39	
MAX	4.15	1.98	4.15	2.98	2.59	3.36	3.69	

TABLE S11: Binding energies (kcal/mol) for the L14 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using five composite electronic-structure methods and three semi-empirical methods.

	L14	HF-3c	PBEh-3c	B97-3c	r ² SCAN-3c	ωB97X-3c	GFN2-xTB	DFTB3-D3H5	PM6-D3H4
C2C2PD	-22.12	-17.47	-20.57	-19.36	-19.43	-20.87	-20.02	-16.95	
C3A	-16.74	-14.85	-16.27	-15.79	-16.72	-15.95	-15.73	-15.37	
C3GC	-27.91	-24.50	-27.84	-27.27	-27.44	-26.75	-27.73	-25.63	
CBH	-12.98	-12.06	-10.58	-9.66	-8.71	-8.70	-13.91	-10.36	
GCGC	-14.47	-13.51	-13.65	-15.42	-15.38	-17.09	-15.87	-19.42	
GGG	-1.03	-1.44	-2.04	-2.84	-2.44	-1.02	-0.76	-3.23	
PHE	-23.26	-23.91	-23.80	-24.44	-25.10	-22.62	-27.14	-25.50	
2a	-36.25	-34.85	-35.89	-34.21	-35.00	-29.91	-31.43	-33.08	
2b	-24.30	-21.68	-22.93	-22.26	-22.62	-20.39	-21.23	-20.92	
3b	-29.69	-31.20	-30.89	-34.00	-33.65	-33.30	-31.91	-29.52	
S8-1	-36.33	-39.02	-39.78	-37.82	-38.13	-34.22	-33.39	-30.66	
S8-2	-30.87	-30.46	-30.92	-31.92	-31.67	-29.43	-32.46	-32.21	
Da1	-21.37	-16.04	-18.88	-17.29	-17.01	-19.71	-20.64	-16.71	
Da2	-21.91	-18.00	-19.73	-18.84	-18.42	-18.86	-20.63	-17.53	
MAE	1.40	1.61	0.82	1.00	1.07	1.65	1.52	2.35	
MAX	3.30	3.59	3.23	1.91	2.35	4.24	3.15	5.88	

TABLE S12: Binding energies (kcal/mol) for the L14 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using IPML and CLIFF machine learning potentials and SAPT0 with jun-cc-pVDZ.

	L14	IPML	CLIFF	SAPT0
C2C2PD	-26.49	—	—	-31.58
C3A	-20.36	—	—	-22.59
C3GC	-33.71	—	—	-37.29
CBH	-17.52	-13.75	—	-8.67
GCGC	-14.34	-16.00	—	-12.92
GGG	-3.09	-5.40	—	-0.77
PHE	-26.44	-28.39	—	-28.58
2a	-40.31	-40.64	—	-46.38
2b	-29.39	-23.14	—	-28.82
3b	—	-40.22	—	-39.96
S8-1	-44.01	-43.68	—	-47.20
S8-2	—	-35.32	—	-39.65
Da1	-24.53	—	—	-28.30
Da2	-26.32	—	—	-28.87
MAE	4.76	4.20	—	6.98
MAX	7.46	7.23	—	12.24

TABLE S13: Binding energies and corresponding energy components (in kcal/mol) for the 11 complexes in the vL11 dataset, computed using SAPT0+aiD(CCD)/aug-cc-pVTZ. The interaction character is designated as “D” (dispersion-dominated, $|E_{\text{disp}}/E_{\text{elst}}| > 2$), “E” (electrostatics-dominated, $|E_{\text{disp}}/E_{\text{elst}}| < 0.5$), or “M” (mixed character, intermediate $|E_{\text{disp}}/E_{\text{elst}}|$).

	vL11	E_{elst}	E_{exch}	E_{ind}	E_{disp}	E_{total}	$ E_{\text{disp}}/E_{\text{elst}} $	Type
	3a	-23.82	55.81	-8.37	-66.40	-42.78	2.79	D
	4a	-33.91	76.20	-7.63	-74.09	-39.43	2.18	D
	5a	-50.66	59.28	-19.24	-42.98	-53.59	0.85	M
	5b	-34.93	46.76	-12.15	-37.64	-37.95	1.08	M
	CiM-a	-65.99	68.75	-23.52	-29.86	-50.62	0.45	E
	CiM-c	-6.79	13.34	-1.83	-20.93	-16.21	3.08	D
	CiM-d	-103.21	110.43	-38.35	-62.01	-93.15	0.60	M
	CiM-e	-88.63	117.09	-28.82	-86.26	-86.62	0.97	M
DNA-ellipticine		-23.02	54.90	-6.68	-68.50	-43.30	2.98	D
C ₆₀ -C ₆₀		-8.38	19.84	-2.37	-16.35	-7.25	1.95	M
C ₆₀ @[6]CPPA		-57.24	128.44	-14.85	-96.99	-40.64	1.69	M

TABLE S14: Binding energies (kcal/mol) for the vL11 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using MP2-based methods. All MP2 calculations, except for MP2+*aiD*(CCD), were extrapolated to the CBS limit, with MP2+*aiD*(CCD) using the aug-cc-pVTZ basis set. BSSE corrections were applied to all MP2 calculations using the counterpoise method.

	vL11	MP2	SCS-MP2	SCS(MI)-MP2	MP2D	SCS-MP2D	MP2+ <i>aiD</i> (CCD)
	3a	-57.30	-39.37	-43.78	-43.85	-42.89	-40.76
	4a	-79.85	-54.97	-60.21	-55.38	-52.31	-38.73
	5a	-48.60	-38.67	-43.83	-44.70	-45.11	-43.91
	5b	-36.32	-27.14	-31.56	-31.76	-31.94	-30.01
	CiM-a	-42.20	-35.95	-40.20	-40.97	-41.28	-39.69
	CiM-c	-19.73	-14.25	-15.71	-16.26	-16.10	-15.22
	CiM-d	-76.21	-63.58	-71.44	-74.92	-75.33	-71.47
	CiM-e	-74.72	-55.81	-64.83	-67.80	-68.42	-64.49
DNA-ellipticine		-55.60	-37.90	-41.90	-42.15	-41.31	-40.67
	C ₆₀ -C ₆₀	-20.22	-13.87	-15.18	-14.62	-13.33	-7.51
C ₆₀ @[6]CPPA		-90.92	-59.11	-65.46	-58.11	-54.39	-37.57
	MAE	15.34	6.77	6.44	5.29	4.49	1.61
	MAX	50.73	18.92	25.27	17.92	14.20	2.75

TABLE S15: Binding energies (kcal/mol) for the vL11 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using selected DFT functionals developed by the Head-Gordon group. All DFT calculations were performed with the def2-TZVPPD basis set and applied counterpoise corrections.

	vL11	B97M-V	B97M-rV	ω B97X-D	ω B97X-V	ω B97M-V
3a	-42.06	-44.25	-39.77	-41.87	-44.09	
4a	-38.48	-40.66	-40.34	-39.50	-43.58	
5a	-42.60	-43.59	-44.41	-43.59	-45.32	
5b	-30.39	-31.34	-30.08	-30.22	-31.59	
CiM-a	-40.70	-41.19	-41.24	-40.38	-40.88	
CiM-c	-15.45	-16.17	-16.82	-16.17	-16.53	
CiM-d	-75.37	-76.77	-75.08	-74.75	-76.59	
CiM-e	-72.09	-74.53	-68.58	-71.35	-73.85	
DNA-ellipticine	-41.34	-43.63	-42.57	-42.27	-43.81	
C ₆₀ -C ₆₀	-8.17	-8.69	-7.74	-8.07	-8.92	
C ₆₀ @[6]CPA	-36.19	-38.65	-37.89	-38.34	-44.65	
MAE	2.01	2.86	1.80	2.00	3.53	
MAX	5.71	8.15	4.19	4.97	7.47	

TABLE S16: Binding energies (kcal/mol) for the vL11 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using selected Minnesota functionals and functionals with D4 and MBD dispersion corrections. All DFT calculations were performed with the def2-TZVPPD basis set and applied counterpoise corrections. The three-body Axilrod–Teller–Muto effects were accounted for in the D4 dispersion correction but not in D3 in this work.

	vL11	B3LYP+D4	PBE0+D4	PBE0+MBD	PBE+D4	M06L-D3	M06-2X-D3	MN15
	3a	-42.00	-36.04	-34.96	-35.23	-39.91	-40.93	-37.45
	4a	-43.37	-37.27	-31.45	-34.94	-32.05	-35.83	-29.91
	5a	-43.12	-42.70	-43.98	-40.17	-40.84	-45.50	-40.10
	5b	-31.06	-29.53	-30.09	-28.32	-28.38	-30.74	-26.52
	CiM-a	-43.35	-43.07	-43.79	-42.56	-40.94	-40.51	-35.60
	CiM-c	-15.37	-14.84	-15.60	-14.46	-15.13	-15.60	-14.27
	CiM-d	-74.54	-72.96	-76.94	-70.36	-78.42	-76.88	-68.45
	CiM-e	-68.88	-64.81	-67.69	-61.55	-73.39	-73.02	-64.03
DNA-ellipticine		-43.43	-37.48	-36.49	-36.48	-40.20	-39.40	-37.40
	C ₆₀ -C ₆₀	-9.48	-8.50	-6.68	-8.42	-5.24	-6.24	-3.48
C ₆₀ @[6]CPA		-44.82	-38.01	-33.44	-35.63	-30.04	-37.61	-29.64
	MAE	2.42	1.05	2.86	2.49	3.65	2.54	4.19
	MAX	5.06	2.18	7.08	4.83	10.14	6.64	10.55

TABLE S17: Binding energies (kcal/mol) for the vL11 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using five composite electronic-structure methods and three semi-empirical methods.

	vL11	HF-3c	PBEh-3c	B97-3c	r ² SCAN-3c	ωB97X-3c	GFN2-xTB	DFTB3-D3H5	PM6-D3H4
	3a	-37.03	-36.42	-37.77	-40.51	-37.94	-38.40	-39.15	-36.64
	4a	-38.34	-36.59	-36.07	-33.38	-35.70	-37.51	-34.70	-30.21
	5a	-39.08	-39.30	-40.25	-42.50	-41.81	-35.47	-40.82	-41.65
	5b	-27.70	-24.59	-27.89	-29.20	-28.83	-28.05	-30.82	-31.56
	CiM-a	-43.69	-37.95	-40.77	-40.18	-41.01	-41.90	-49.09	-49.31
	CiM-c	-14.76	-14.29	-14.86	-15.15	-14.70	-13.00	-13.96	-13.26
	CiM-d	-69.36	-65.64	-66.71	-70.06	-70.43	-64.55	-86.89	-79.88
	CiM-e	-66.30	-56.38	-59.13	-65.52	-63.40	-69.01	-78.07	-76.91
DNA-ellipticine		-40.42	-35.64	-37.80	-39.48	-37.68	-40.02	-38.52	-38.85
C ₆₀ -C ₆₀		-7.67	-7.30	-7.60	-7.60	-8.10	-10.90	-7.67	-6.57
C ₆₀ @[6]CPPA		-42.10	-38.61	-36.78	-33.80	-37.77	-46.49	-37.19	-29.29
MAE		1.75	3.66	2.51	2.14	1.42	3.05	4.25	4.86
MAX		4.33	10.00	7.26	6.39	3.25	9.14	13.20	10.90

TABLE S18: Binding energies (kcal/mol) for the vL11 dataset, including mean absolute error (MAE) and maximum absolute error (MAX), calculated using IPML and CLIFF machine learning potentials and SAPT0 with jun-cc-pVDZ.

	vL11	IPML	CLIFF	SAPT0
	3a	-43.44	-48.55	-46.06
	4a	-47.72	—	-69.88
	5a	-46.90	-61.39	-48.21
	5b	-37.62	-44.62	-35.96
	CiM-a	-46.84	-58.26	-44.38
	CiM-c	-17.79	-14.54	-16.43
	CiM-d	-73.87	-83.01	-82.50
	CiM-e	-77.32	-77.26	-76.10
DNA-ellipticine	—	—	—	-45.00
	C ₆₀ -C ₆₀	-10.36	—	-17.97
	C ₆₀ @[6]CPPA	-46.77	—	-77.11
	MAE	5.40	11.67	11.51
	MAX	10.93	18.67	36.92

TABLE S19: Summary of binding mean absolute error (MAE) in kcal/mol for all computational methods tested in this work on the L14 and vL11 datasets, along with the MAE differences between the two. The difference is highlighted in red if its absolute value is ≤ 0.5 kcal/mol, and in blue if its absolute value is > 0.5 but ≤ 1.0 kcal/mol.

method	MAE _{L14}	MAE _{vL11}	MAE _{vL11} - MAE _{L14}
MP2	11.26	15.34	4.08
SCS-MP2	2.59	6.77	4.18
SCS(MI)-MP2	4.54	6.44	1.90
MP2D	3.32	5.29	1.97
SCS-MP2D	2.89	4.49	1.60
MP2+aiD(CCD)	1.24	1.61	0.37
B3LYP+D4	2.29	2.42	0.13
PBE0+D4	0.68	1.05	0.37
PBE0+MBD	1.81	2.86	1.05
PBE+D4	1.24	2.49	1.25
B97M-V	0.65	2.01	1.36
B97M-rV	1.64	2.86	1.22
ω B97X-V	1.39	2.00	0.61
ω B97M-V	2.24	3.53	1.29
ω B97X-D	1.90	1.80	-0.10
B97M-V+E ^{ABC}	0.74	1.65	0.91
B97M-rV+E ^{ABC}	0.37	1.91	1.54
ω B97X-V+E ^{ABC}	0.76	1.40	0.64
ω B97M-V+E ^{ABC}	0.99	1.80	0.81
ω B97X-D+E ^{ABC}	1.11	1.29	0.18
M06L-D3	0.99	3.65	2.66
M06-2X-D3	1.04	2.54	1.50
MN15	1.39	4.19	2.80
HF-3c	1.40	1.75	0.35
PBEh-3c	1.61	3.66	2.05
B97-3c	0.82	2.51	1.69
r ² SCAN-3c	1.00	2.14	1.14
ω B97X-3c	1.07	1.42	0.35
GFN2-xTB	1.65	3.05	1.40
DFTB3-D3H5	1.52	4.25	2.73
PM6-D3H4	2.35	4.86	2.51
IPML	4.76	5.40	0.64
CLIFF	4.20	11.51	7.31
SAPT0	6.98	11.67	4.69

TABLE S20: Summary of binding maximum absolute error (MAX) in kcal/mol for all computational methods tested in this work on the L14 and vL11 datasets, along with the MAX differences between the two. The difference is highlighted in red if its absolute value is ≤ 0.5 kcal/mol, and in blue if its absolute value is > 0.5 but ≤ 1.0 kcal/mol.

method	MAX _{L14}	MAX _{vL11}	MAX _{vL11} - MAX _{L14}
MP2	18.08	50.73	32.65
SCS-MP2	5.68	18.92	13.24
SCS(MI)-MP2	8.52	25.27	16.75
MP2D	5.85	17.92	12.07
SCS-MP2D	5.22	14.20	8.98
MP2+ <i>ai</i> D(CCD)	2.60	2.75	0.15
B3LYP+D4	4.15	5.06	0.91
PBE0+D4	1.98	2.18	0.20
PBE0+MBD	4.15	7.08	2.93
PBE+D4	2.98	4.83	1.85
B97M-V	1.57	5.71	4.14
B97M-rV	3.16	8.15	4.99
ω B97X-V	4.47	4.97	0.50
ω B97M-V	4.47	7.47	3.00
ω B97X-D	4.48	4.19	-0.29
B97M-V+E ^{ABC}	1.47	6.95	5.48
B97M-rV+E ^{ABC}	1.50	5.26	3.76
ω B97X-V+E ^{ABC}	2.63	4.80	2.17
ω B97M-V+E ^{ABC}	2.63	4.58	1.95
ω B97X-D+E ^{ABC}	2.92	5.24	2.32
M06L-D3	2.59	10.14	7.55
M06-2X-D3	3.36	6.64	3.28
MN15	3.69	10.55	6.86
HF-3c	3.30	4.33	1.03
PBEh-3c	3.59	10.00	6.41
B97-3c	3.23	7.26	4.03
r ² SCAN-3c	1.91	6.39	4.48
ω B97X-3c	2.35	3.25	0.90
GFN2-xTB	4.24	9.14	4.90
DFTB3-D3H5	3.15	13.20	10.05
PM6-D3H4	5.88	10.90	5.02
IPML	7.46	10.93	3.47
CLIFF	7.23	18.67	11.44
SAPT0	12.24	36.92	24.68